Team 7

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I. Study Protocols

a. Heuristic Evaluation

A heuristic evaluation was conducted using four expert evaluators to assess the usability of the interface. The team used four evaluators based on a 1994 paper in which Jakob Nielsen, the designer of heuristic principles, recommended 3-5 evaluators in order to ensure the evaluation is thorough and accurate [1]. Each evaluator independently reviewed the prototype based on Nielsen's 10 heuristics and identified violations. Evaluators were instructed to focus on identifying violations of usability principles rather than simply confirming whether the interface met each heuristic.

Step-By-Step Procedures.

- Gather four experts for the heuristic evaluation.
- Inform evaluators of Nielsen's ten heuristic principles.
- Instruct evaluators to independently interact with the prototype, reviewing all of the major tasks assigned and screens.
- While navigating the interface, instruct evaluators to identify usability issues corresponding to those heuristics.
- For each issue, have evaluators record observations using the provided Google Form.
- Collect all responses and later aggregate them to identify common usability concerns, prioritizing improvements and next steps.

Materials Used.

- Designer's laptop
- "Phish or Fail" prototype
- Evaluation Google Form
- Instruction sheet (which includes the list of Nielsen's 10 usability heuristics)

Scripts.

Refer to *Material B* to see the script used for the heuristic evaluation.

Methods of Data Recording.

- Data was collected using a standardized Google Form to ensure consistency across evaluators. For each usability issue identified, evaluators recorded the following:
    - Heuristic Violated: Selected from a dropdown list of Nielsen's 10 heuristics
    - Description of the Issue: A clear explanation of the problem
    - Location of the Issue: Specific screen, task, or feature where the issue occurred
    - Severity Rating (0–4):
        - 0 = Not a problem
        - 1 = Cosmetic issue
        - 2 = Minor usability problem
        - 3 = Major usability problem
        - 4 = Usability catastrophe
    - Suggested Improvement (optional): Recommendations for resolving the issue

- Responses gathered were automatically uploaded to a Google Sheets spreadsheet.

b. Predictive Evaluation

To evaluate the efficiency of the interface, a predictive evaluation was conducted using the Keystroke-Level Model (KLM). KLM is used to estimate how long it takes a user to complete a task by breaking it down into smaller actions such as mental preparation, pointing, and clicking. This method allows developers to assess usability without testing users in real-time. Experts, in this case, developers, assign times to the broken down tasks and calculate predicted time to assess usability [2].

Step-By-Step Procedures.

- View task list
- Break down each individual task into smaller actions and steps
- Assign KML timing values to each smaller action
- Calculate total predicted time

Materials Used.

- Designer's laptop
- "Phish or Fail" prototype
- KLM reference sheet
- Task decomposition sheet

Scripts.

A script was not utilized for the predictive (Keystroke-Level Model) evaluation.

Methods of Data Recording.

- Data was recorded using a task sheet and manually entered onto a spreadsheet with task number and predicted time-on-task column (*Material D, Material E*)

c. Think-Aloud Evaluation

In Think-Aloud Evaluations, users interact with the prototype while voicing all of their inner thoughts and feelings in order to give in-depth insight on how users are choosing to interact with the system.

Step-By-Step Procedures.

- Recruit 5 new participants who are not enrolled in CPSC4140 or have previous experience with the application.
- Ensure the user is sitting down with the provided laptop, and have a single tab open to the "Phish or Fail" application.
- The administrator first obtains consent by reading the overview and consent script and then begins recording the audio, if consent is given.
- The user is given a list of tasks to complete as they navigate the application.

- The administrator reads from the instructions script and prompts the user for final questions before beginning the evaluation.
- The user will complete the lists of tasks, and the administrator remains silent as they verbalize their thought process. If the user stops speaking, the administrator prompts them to start sharing their thoughts and what they are doing again.
- Once the user is finished, the administrator thanks them for their time and asks for any closing thoughts before ending the audio recording.
- The audio recording is then transcribed for further review.
- Ask the user to complete the questionnaire after the conclusion of the evaluation.

Materials Used.

- Designer's laptop
- "Phish or Fail" prototype
- Small lists of tasks for user (given through an instruction sheet)
    - Complete a scenario as the Target and share your results verbally.
    - Complete a scenario as the Attacker and share your results verbally.
    - Find the Learning page.
    - Find the Setting page and change the difficulty level to Hard.
    - Complete a hard Target scenario using the hints.
- Recording device for audio
- Document for transcribing audio
- Questionnaire (Google Form)

Scripts.

Refer to *Material G* to see the script used for the think-aloud evaluation.

Methods of Data Recording.

- Audio recording of user's thought process
- Transcript made of recording for parsing and observing trends, patterns, and issues

d. Usability Testing

In usability evaluation, participants are given a task to complete within the prototype and time-on-task, as well as errors and completion, are recorded by evaluators.

Step-By-Step Procedures.

- Recruit 5 new participants who are not enrolled in CPSC4140 or have previous experience with the application.
- Ensure the user is sitting down with the provided laptop, and have a single tab open to the "Phish or Fail" application.
- The administrator should have a Google Sheets spreadsheet to record the completion times for each of the user's benchmark tasks.
- Present the user with a printed out sheet of the 10 benchmark tasks (printed front and back of the page) to read.

- The administrator will then read from the introduction script, highlighting the directions verbally to the user.
- Once any questions are answered and both the administrator and user are ready to begin completing the tasks, the following is completed for each task:
    - The administrator will confirm the task about to be completed (ex. "Now beginning Task 1")
    - The administrator will say "Go!" and start the stopwatch on their phone.
    - The user will follow the steps on the instruction sheet for the assigned task and complete it to the best of their ability.
    - When the user has completed the last step and finishes the task, the administrator stops the stopwatch on their phone.
    - The administrator records the time in the Google Sheets spreadsheet under the correct task and user cell.
- When all 10 benchmark tasks are completed the administrator will thank the user for their time and the evaluation will end.
- Ask the user to complete the questionnaire after the conclusion of the evaluation.

Materials Used.

- Designer's laptop
- "Phish or Fail" prototype
- Instruction sheet
- Device with a stopwatch application
- Google Sheets spreadsheet for data collection
- Questionnaire (Google Form)

Scripts.

Refer to *Material J* to see the script used for the usability evaluation.

Methods of Data Recording.

- A stopwatch to record the time taken for each benchmark task
- Google Sheets spreadsheet for data collection, including time, success, and notes

e. Questionnaire

Participants in the Think-Aloud and Usability evaluations complete a questionnaire about the prototype after the study.

Step-By-Step Procedures.

- After completing the think-aloud evaluation or the usability evaluation, the ten participants should be directed to complete a post-questionnaire.
- Send the questionnaire via Google Forms, ensuring all questions are included, and responses are collected anonymously.
- Instruct the 10 participants to complete the survey immediately after their evaluation.
- Once all responses are collected, export the answers to a spreadsheet.

- Use the responses to identify trends, compute any quantitative responses and group common themes identified in open-response questions.

Materials Used.

- Participant's/Designer's laptop
- "Phish or Fail" prototype
- Questionnaire (Google Form)
- Response spreadsheet

Scripts.

Refer to *Material M* for the script used for questionnaire administration.

Methods of Data Recording.

- Responses were collected through Google Forms
- After all responses were recorded, the data were exported to a spreadsheet
- Quantitative data was analyzed by calculating averages and frequency
- Qualitative data was analyzed through theme extraction and grouping
- Results were organized and shown through tables and charts

II. Participant Demographics

Participants in the various evaluations were all college students at Clemson University, aged 18 to 22. They were of varying ethnicities, including Caucasian, Asian, African American, and more. All participants had basic computer literacy and described themselves as having basic to intermediate knowledge of financial scams, including first-hand experience as targets. Importantly, all participants had no prior knowledge of the prototype development, ensuring that no prior biases were present.

III. Task Design

a. Tasks Evaluated

For the evaluation of "Phish or Fail," the team utilized a set of ten benchmark tasks for participants to work through. These tasks are listed here:

1. Begin Game
2. Select Role and Scenario
3. Complete Target Scenario
4. Receive Score and View Feedback (Target)
5. Complete Scammer Scenario
6. Receive Score and View Feedback (Scammer)
7. Navigate to Next Action
8. Navigate to Learning Page
9. Use Hints to Complete Target Scenario
10. Navigating to Settings and Change Difficulty

For heuristic evaluations, no specific tasks were given, as the goal was to have evaluators move through the system naturally and extensively. In predictive evaluation using the KLM method, those ten tasks were broken down into smaller pieces. That breakdown is available in *Material D.* For the Think-Aloud evaluation, the above task list was shorted and modified to:

1. Complete a scenario as the Target and share your results verbally.
2. Complete a scenario as the Attacker and share your results verbally.
3. Find the Learning page.
4. Find the Setting page and change the difficulty level to Hard.
5. Complete a hard Target scenario using the hints.

b. Selection of Tasks Over Alternatives

Heuristic Evaluation. The evaluators were not given specific tasks for the heuristic evaluation. They were instructed to freely explore the prototype and act naturally. This approach was chosen to better reflect natural user behavior and allow evaluators to interact with the system in a more realistic manner. By navigating the interface with no tasks, evaluators were able to encounter violations of heuristic design problems that might not be evident while completing specific tasks. Although no tasks were assigned, the design of the system naturally guides evaluators through key functionality areas, including starting the game, selecting roles, completing scenarios, receiving feedback, and accessing the learning page and settings. This ensured that all major features of the application were still examined. Compared to a structured task list, this method allowed for broader coverage of the system, which increases the diversity of heuristics violations identified.

Think-Aloud Evaluation. The tasks given to users for this evaluation covered the full navigation and interaction with the prototype. It provides a full user experience, from starting the game to engaging with different scenarios and exploring features. Compared with the predictive and usability evaluation benchmark tasks (discussed later), the tasks were shortened and simplified to reduce users' cognitive load. They were described succinctly, highlighting the core action, and users were able to focus more on thinking aloud and expressing their thoughts and opinions. An abundance of overly complicated tasks could have hindered data collection.

Predictive and Usability Evaluation. The KLM and usability evaluations used the same 10 consistent benchmarks, allowing the system to be evaluated from the perspectives of both an expert and a user. For KLM, tasks are broken down into operators that provide quantitative benchmarks. In comparison, the usability evaluation focuses on qualitative and quantitative benchmarks, additionally capturing user behavior and options. Similar to the other evaluations, these tasks also assessed system functionality rather than independent and irrelevant feature testing. Overall, unlike other potential tasks, these allowed for efficiency measurement (from both an expert and users), user behavior observations, and a comprehensive system coverage.

c. Replicability

To allow others to reproduce the studies, Part I, Design Protocols, includes step-by-step procedures, materials used, and data recording methods. The appendix contains materials *A*-*N*, which include the instruction sheet, scripts, and raw data collected. All of the evaluations are easily reproducible due to their consistency, standardization, and similar benchmark tasks. After the think-aloud and usability evaluations, the ten participants completed a questionnaire, providing insight into their usability experience through multiple-choice, ranking, Likert-scale, and other question types. The questions asked within the questionnaire were chosen with intention and purpose. Distinguishing which study the user participated in allowed comparison of the results between the two. The prototype's navigation, consistency, and visibility of system status were evaluated. The questionnaire also determined whether the system matched real-world conditions, mimicking realistic scenarios. Additionally, it captured the user's background and the occurrence of exposure, which helped determine the varying performance amongst the participants. The questions also measured the learning outcomes and system effectiveness. Ranking the features also identified the factors users considered most contributing. The open-ended questions allowed users to provide input on specific issues they identified and suggestions for improving future iterations of the system prototype.

IV. Results and Analysis

a. Results Summary

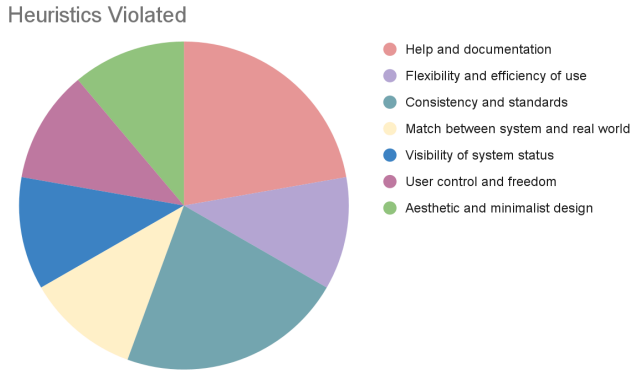

Figure 1.1: Heuristic Evaluation Results

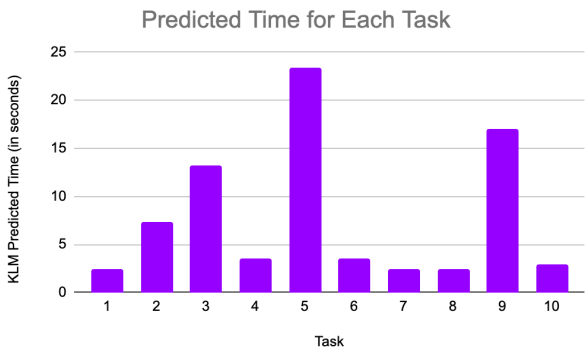

Figure 1.2: Predictive Evaluation Results

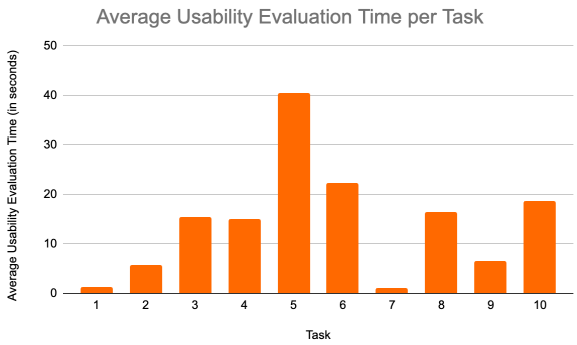

Figure 1.3: Usability Evaluation Results

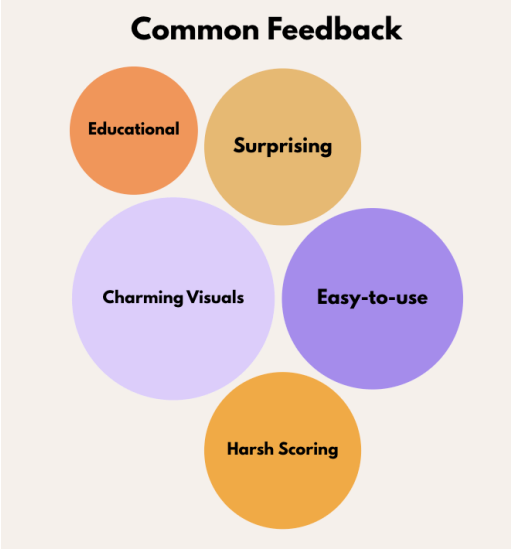

Figure 1.4: Think-Aloud Common Themes

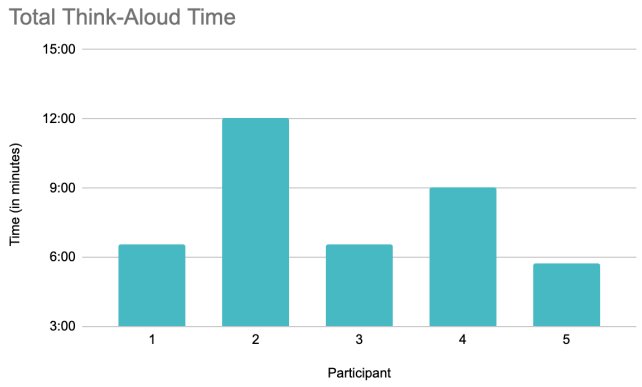

Figure 1.5: Think-Aloud Run Times

Discussion

Overall, our team has concluded that the evaluations demonstrate that our product is usable and aligned with human-centered design principles. As seen in Figure 1.4, participants with no prior knowledge of the prototype described it as educational and easy to use, which is favorable feedback. Additionally, during the heuristic evaluation, expert evaluators did not encounter any severe or inhibiting issues; instead, they noted small design features that should be addressed. Functionality was not an issue in the heuristic evaluations. The predictive and usability evaluation results show that the design does not place a high cognitive load on users, as participants in the usability studies did not average notably high times on any of the tasks. The highest average time was 40 seconds, indicating that the tasks are not too burdensome for the user. The predictive user times were slightly lower than the actual usability times for some of the more involved tasks, such as working through a scenario, which is also good data for our team. This discrepancy will be helpful as we think about the next design phase, as it suggests we need to improve readability to increase user interaction ease.

The biggest issue we noticed was readability: in text-heavy scenarios, users struggled to remain engaged and fully comprehend what was being asked of them. When working on the next iteration of this prototype, our team's goal will be to adjust the length of both scenarios and feedback. One initial idea we have is to incorporate progressive disclosure to reduce cognitive load and avoid overwhelming the player with information all at once.

b. Synthesis

The design of our prototype is effective in achieving its educational game goal; however, both the heuristic evaluation and think-aloud findings highlight areas for improvement in usability and user engagement. Heuristic evaluators noted that certain design principles could be strengthened, particularly regarding text readability and button affordances, to make the interface more intuitive and visually engaging. Additionally, feedback from the think-aloud sessions revealed that users

were less willing to engage with large blocks of text and experienced frustration when receiving incorrect answers without sufficient feedback or opportunity for more hints. These insights suggest that while the core concept is strong, the interface can be further refined to reduce cognitive load, improve clarity, and create a more supportive user experience. Enhancing these elements will better align the system with user-centered design principles and ultimately make the learning experience more effective and enjoyable.

Despite the identified usability challenges, the prototype largely meets its core goals outlined in Milestone 3. The system effectively improves phishing awareness and user knowledge by allowing users to actively practice identifying scams rather than passively consuming information. It also successfully provides a safe, low-stakes learning environment, where users can make mistakes without real-world consequences, which is critical for building confidence. In a post-evaluation questionnaire, users reported that the design was both navigable and realistic. The interactive, game-based format contributes to an engaging experience, as users spoke positively about it in both think-aloud evaluations and the post-evaluation questionnaire. However, the usability issues identified (text-heavy content and limited feedback) slightly hinder the overall experience, particularly in terms of engagement and ease of use. With specific improvements in the next design iteration, the system has high potential to fully achieve all its intended goals.

c. Roadmap

After reviewing the results, our team created a plan for further enhancements to new implementations and to “Phish or Fail.” More scenarios would be implemented for each category, so a user can have greater variability at each level of difficulty. In addition to adding more scenarios for each level, another implementation would be to add an adaptive difficulty aspect to the game. This will allow users to naturally target their weak areas and improve their knowledge through a more individually tailored process as they work through scenarios at each level of difficulty. Another improvement would be to break up the blocks of text and add a visual hierarchy using different headings to make it easier for users to digest, read and understand their feedback on scenarios. Next, an implementation that could be useful for users is progress tracking to see their stats for each level of difficulty, and possibly work towards a goal of getting 10 easy, 10 medium, and 10 hard scenarios correct to earn a badge, which can help motivate them to continue engaging. There should also be a fully accessible system implementation so that users who need text-to-speech, need the text to be larger, or need a color scheme changed to see can have these options. This could be added to settings or have its own section, so users who need these features have a one-stop shop to find the tools. Finally, animated visuals could continue to grab attention, engage a younger audience and improve gameplay, providing an extra boost to keep users' attention on the game “Phish or Fail.”

V. Appendix

Material A. Heuristic Evaluation Instruction Sheet

Material B. Heuristic Evaluation Script
"Thank you for participating in this heuristic evaluation. The purpose of this study is to assess the usability of our application by identifying potential design issues based on established usability principles from Jakob Nielsen's 10 usability heuristics. As you interact with the prototype, please focus on identifying any usability problems or areas where the design does not align with these principles. For each issue you find, please record your observations using the provided Google Form. There are no right or wrong answers - our goal is to gather as many insights as possible. Please work independently and take your time exploring all relevant parts of the interface. Thank you again for your participation."

Material C. Heuristic Evaluation Raw Data

Material D. Predictive Evaluation Task Breakdown

Material E. Predictive Evaluation Raw Data

Material F. Think-Aloud Evaluation Instruction Sheet

Material G. Think-Aloud Evaluation Script
"Thank you for your time today. We are conducting research about how users interact with our web application and we are interested in seeing how you interact with it. The application is a simple educational game that seeks to teach users about scam messages. Participation in this evaluation will involve completely some set of tasks and sharing your thought process as you move through them and should take about 5-10 minutes. Your name will not be released. Notes will be taken on what you say throughout the process. Would it be okay if we audio record you as you go? You may stop whenever you like but we will use whatever data collected before that point. Ready to begin? **Obtain consent before continuing and then begin recording before reading aloud the instructions** You are participating in what is called a "think aloud" evaluation. As you see and do anything, share your thoughts as they come and describe your actions as well, simply think aloud. It might feel odd, but try your best to vocalize your stream of consciousness as you play Phish or Fail. We have a small list of tasks for you to complete for this evaluation which is in the list I have given to you. Are you ready?"

Material H. Think-Aloud Evaluation Raw Data

Material I. Usability Evaluation Instruction Sheet

Material J. Usability Evaluation Script
"Thank you for participating in the Usability Evaluation! Today I'm going to be administering this evaluation and I'll walk to you through the directions and each benchmark task before we start. First, please read over all 10 of the benchmark tasks on the paper provided. One major rule of this evaluation is to not start a new benchmark task without my signal "Go!" I will be timing you and for these times to be accurate I need to start/stop it when you begin and end each task. Any questions before we start?"

Material K. Usability Evaluation Raw Data

Material L. Questionnaire Google Form

Material M. Questionnaire Script
"Thank you for taking the time out of your day to participate in our think-aloud/usability session. Please take a few minutes to complete this survey about your experience with the "Phish or Fail" prototype. Your responses are anonymous and appreciated. Answer each question as honestly as possible and to the best of your ability."

Material N. Questionnaire Raw Data

References

[1] Jakob Nielsen. 1994. The Theory behind Heuristic Evaluations. *Nielsen Norman Group*. Retrieved from https://www.nngroup.com/articles/how-to-conduct-a-heuristic-evaluation/theory-heuristic-evaluations/

[2] Zbynek Sterba. 2017. Keystroke Level Modeling as a Method for Usability Quantification. *Medium*. Retrieved from https://medium.com/@zbynekste/keystroke-level-modeling-as-a-method-for-usability-quantification-8d2a390f69da